

Artificial Intelligence

For HEDSPI Project

Lecture 2 - Agent

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Outline

1. Agents and environments
2. PEAS (Performance measure, Environment, Actuators, Sensors)
3. Environment types
4. Agent types

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Agents and environments

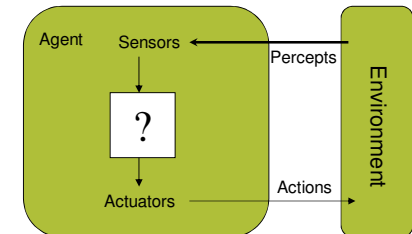
- An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and **acting** upon that environment through **actuators**
- Example 1: human agent
 - Sensors: eyes, ears, ...
 - Actuators: hands, legs, mouth, ...
- Example 2: robotic agent (e.g., Aishimo)
 - Sensors: camera, infrared range finders
 - Actuators: various motors

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Agents and environments (con't)

- The **agent function** maps from percept histories to actions:

$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$



- The **agent program** runs on the physical **architecture** to produce the agent function
agent = architecture + program

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Agent function based on conditional table

Function TABLE-DRIVEN-AGENT(*percept*) **returns** an action

static: *percepts*, a sequence, initially empty

table, a table of actions, indexed by percept sequences, initially fully specified

Append *percept* to the end of *percepts*

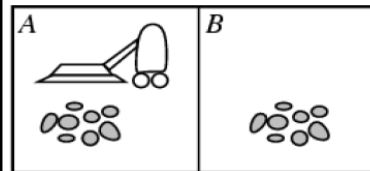
$action \leftarrow \text{LOOKUP}(percepts, table)$

Return *action*

Drawback: huge table!

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Vacuum-cleaner world



- Percepts: location (A or B), state (clean or dirty)

- Actions: *Left*, *Right*, *Suck*, *NoOp*

Percept sequence	Action
[A, clean]	Right
[A, dirty]	Suck
[B, clean]	Left
[B, dirty]	Suck
[A, clean][A, clean]	Right
[A, clean][A, dirty]	Suck

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Vacuum-cleaner world

Function Reflex-Vacuum-Agent([position, state]) **returns**
action

If state = Dirty **then return** Suck

Else if position = A **then return** Right

Else if position = B **then return** Left

End Function

- Does the agent act reasonably?

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Rational agent

- A **rational agent** is one that does the right thing - the one that will cause the agent to be most successful

- **Performance measure** embodies the criterion for success of an agent's behavior.

□ E.g., performance measure of a vacuum-cleaner agent:

- amount of dirt cleaned up
- amount of time taken
- amount of electricity consumed
- amount of noise generated
- ...

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Rational agent

- For each possible percept sequence, a rational agent should select an **action** that is expected to maximize its **performance measure**, given the evidence provided by the **percept sequence** and whatever **built-in knowledge** the agent has.
- An agent is **autonomous** if its behavior is determined by its own experience (with ability to learn and adapt)

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PEAS

- 4 factors should be considered when design an automated agent:
 - **P**erformance measure
 - **E**nvironment
 - **A**ctuators
 - **S**ensors

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PEAS - automated taxi driver

- **Performance measure:** Safe, fast, legal, comfortable trip, maximize profits, ...
- **Environment:** Roads, other traffic, pedestrians, weather, ...
- **Actuators:** Steering wheel, accelerator, brake, signal, horn, ...
- **Sensors:** Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard, ...

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PEAS - Medical diagnosis system

- **Performance measure:** Healthy patient, minimize costs, lawsuits, ...
- **Environment:** Patient, hospital, staff
- **Actuators:** Screen display (questions, tests, diagnoses, treatments, referrals)
- **Sensors:** Keyboard (entry of symptoms, findings, patient's answers)

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PEAS - Spam Filtering Agent

- **Performance measure**: spam block, false positives, false negatives
- **Environment**: email client or server
- **Actuators**: mark as spam, transfer messages
- **Sensors**: emails (possibly across users), traffic, etc.

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Environment types

- **Fully observable** (vs. partially observable): An agent's sensors give it access to the complete state of the environment at each point in time.
- **Deterministic** (vs. stochastic): The next state of the environment is completely determined by the current state and the action executed by the agent.
- **Episodic** (vs. sequential): The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action).

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Environment types

- **Static** (vs. dynamic): The environment is unchanged while an agent is deliberating.
- **Discrete** (vs. continuous): A limited number of distinct, clearly defined percepts and actions.
- **Single agent** (vs. multiagent): An agent operating by itself in an environment.

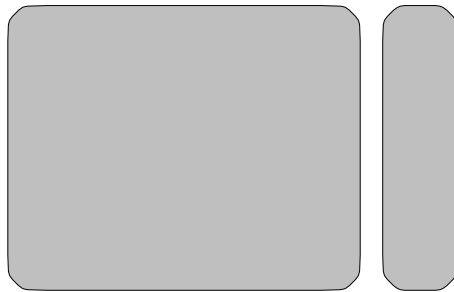
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Agent types

- Four basic agent types:
 - Simple reflex agents
 - Model-based reflex agents
 - Goal-based agents
 - Utility-based agents

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Simple reflex agent



Function SIMPLE-REFLEX- AGENT(*percept*) **returns** an action
static: *rules*, a set of condition-action rules
state $\leftarrow$ INTERPRET-INPUT(*percept*)
rule $\leftarrow$ RULE-MATCH(*state*, *rules*)
action $\leftarrow$ RULE-ACTION[*rule*]
return *action*

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ERROR: rangecheck
OFFENDING COMMAND: get

STACK:

1
[[-173 -307 1097 979]]
0
/descender
[(Agent)]
-savelevel-
-savelevel-